

Project 3 – 5G-connected high-speed train

A high-speed train moves at constant speed V along a railway of length $L = 10\text{km}$. Assume that when the train reaches the end of the railway it “reappears” at the starting point, continuing its motion seamlessly.

The train provides WiFi connectivity to N passengers through an onboard wireless Access Point (AP), and the AP is connected to a 5G radio network consisting of M base stations (BSs) placed alongside the railway at distance $D = L/M$ between each other.

Each passenger generates packets of size S bytes every T seconds, and transmits them to the AP (S and T are IID random variables whose distribution is specified below). In turn, the AP forwards packets to its *closest* BS. The AP can transmit only one packet at a time following a First In First Out (FIFO) policy, at a rate $R = C * e^{-(d/1000)^2}$, where C is the maximum data rate (in bytes/s) and d is the distance between the AP and the BS.

Before starting each transmission, the AP checks which BS is the closest. If the closest BS differs from the current serving BS, the AP needs to perform a *handover* operation that takes H seconds. The AP cannot perform any transmission during the handover and resumes transmissions after the handover is complete.

Evaluate at least the overall system throughput for various values of N and M in the following scenario(s):

- Exponential distribution for both T and S .

In all cases, it is up to the team to calibrate the scenarios so that meaningful results are obtained.

Project deliverables:

- a) Documentation (according to the standards set during the lectures)
- b) Simulator code
- c) Presentation (up to 10 slides maximum)